

Modelling part

2026-05-13

```
# see datacollection.R
# load the main dataframe
df <- read.csv('datasources/df_essay.csv')
```

```
# create a dependent variables for a multinomial model
df_final <- df %>%
  mutate(
    onset_type = case_when(
      `NVC2.1_VIOL` == 1 ~ "Violent",
      `NVC2.1_NONVIOL` == 1 ~ "Nonviolent",
      TRUE ~ "No onset"
    ),
    onset_type = factor(
      onset_type,
      levels = c("No onset", "Nonviolent", "Violent")
    )
  )
```

```
# load functions for multinomial model simulations
source('simulation_functions.R')
```

```
# estimate a two-way FE model (country-year) with lagged independent variables
# to reduce reversed causality problems
# the specifications includes discriminated population share and share of ethnic groups in power
# controls: oilrents, GDP, GDP growth, executive corruption, democracy (quadratic term added),
# regime duration, GINI coefficient (disposable income)
model <- multinom(
  onset_type ~ growup_discrimpop_l + growup_egippop_l +
    log_oilrents_l + log_gdp_pcap_l + pop_log_l + gdp_growth_l +
    v2x_execorr_l + v2x_polyarchy_l + I(v2x_polyarchy_l^2) +
    log_v2regdur_l + swid_gini_disp_l + as.factor(cow) + as.factor(year),
  data = df_final, maxit = 1000 #the default number of iterations is not enough
)
```

```
## # weights: 783 (520 variable)
## initial value 5451.314176
## iter 10 value 1164.557877
## iter 20 value 732.102318
## iter 30 value 694.181232
## iter 40 value 684.044442
## iter 50 value 679.170623
## iter 60 value 676.614146
## iter 70 value 674.945489
```

```
## iter 80 value 674.267434
## iter 90 value 673.485086
## iter 100 value 672.759692
## iter 110 value 672.372600
## iter 120 value 672.328910
## iter 130 value 672.271988
## iter 140 value 672.270800
## iter 150 value 672.261856
## final value 672.261053
## converged
```

```
#print a coefficient tables
```

```
df_raw <- tidy(model, conf.int = FALSE) |>
  dplyr::filter(!str_detect(term, "as.factor")) |>
  dplyr::mutate(
    term = dplyr::recode(term,
      "growup_discrimpop_l" = "Discriminated population share",
      "growup_egipop_l" = "EGI population share",
      "log_oilrents_l" = "Oil rents (log)",
      "log_gdp_pcap_l" = "GDP per capita (log)",
      "pop_log_l" = "Population (log)",
      "gdp_growth_l" = "GDP growth",
      "v2x_execorr_l" = "Executive corruption",
      "v2x_polyarchy_l" = "Polyarchy",
      "I(v2x_polyarchy_l^2)" = "Polyarchy$^{2}$",
      "log_v2regdur_l" = "Regime duration (log)",
      "swid_gini_disp_l" = "Gini (disposable)"
    ),
    stars = dplyr::case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01 ~ "**",
      p.value < 0.05 ~ "*",
      TRUE ~ ""
    ),
    coef = paste0(round(estimate, 3), stars),
    se = paste0("(", round(std.error, 3), ")")
  )
```

```
# Coefficient rows
```

```
coef_rows <- df_raw |>
  dplyr::select(term, y.level, coef) |>
  tidyr::pivot_wider(names_from = y.level, values_from = coef)
```

```
# SE rows - blank term label
```

```
se_rows <- df_raw |>
  dplyr::select(term, y.level, se) |>
  tidyr::pivot_wider(names_from = y.level, values_from = se) |>
  dplyr::mutate(term = "")
```

```
# Interleave: coef row then SE row for each variable
```

```
result <- dplyr::bind_rows(
  coef_rows |> dplyr::mutate(ord = (dplyr::row_number() - 1) * 2 + 1),
  se_rows |> dplyr::mutate(ord = (dplyr::row_number() - 1) * 2 + 2)
) |>
```

```

dplyr::arrange(ord) |>
dplyr::select(-ord)

# Table
result |>
  kable(
    format      = "latex",
    booktabs    = TRUE,
    escape      = FALSE,
    linesep     = "",                                # remove automatic row spacing
    col.names   = c("", "Nonviolent", "Violent"),
    caption     = "Multinomial logit estimates"
  ) |>
  kable_styling(
    latex_options = c("hold_position", "scale_down"),
    font_size     = 10
  ) |>
  footnote(
    general = c(
      "Country and year fixed effects included.",
      "All independent variables are one year lagged",
      "Standard errors in parentheses.",
      "* p<0.05, ** p<0.01, *** p<0.001"
    ),
    escape   = FALSE
  )

```

```

#extract the variance-covariance table
vcov_multinom <- vcov(model)

```

```

# Simulate expected probabilities of non-onsets, violent-nonviolent onsets
# with an observed values approach
# by the values of discriminated population share
# the chunk uses sim_function from the simulation_functions.R
X      <- model.matrix(model)
col_idx <- which(colnames(X) == "growup_discrimpop_1")
K_minus_1 <- nrow(coef(model))
coefs_vec <- as.vector(t(coef(model)))
cat_names <- c(
  levels(df_final$onset_type)[1], # reference category
  rownames(coef(model))           # K-1 non-reference categories
)

xseq <- seq(
  quantile(df_final$growup_discrimpop_1, 0.01, na.rm = TRUE),
  quantile(df_final$growup_discrimpop_1, 0.99, na.rm = TRUE),
  length.out = 20
)

ova_results <- lapply(xseq, function(x_val) {

  # OVA: substitute x_val, keep all other covariates at observed values
  X_mod      <- X

```

Table 1: Multinomial logit estimates

	Nonviolent	Violent
(Intercept)	-28.116*** (2.921)	-94.993*** (2.767)
Discriminated population share	1.186 (2.265)	6.672** (2.261)
EGI population share	-0.835 (0.771)	2.607 (1.608)
Oil rents (log)	0.719* (0.326)	0.523 (0.368)
GDP per capita (log)	0.116 (0.55)	1.878* (0.833)
Population (log)	1.025*** (0.242)	3.344*** (0.504)
GDP growth	-0.012 (0.021)	-0.022 (0.021)
Executive corruption	-2.041* (0.894)	-1.568 (1.448)
Polyarchy	4.324 (3.584)	20.5*** (5.898)
Polyarchy ²	-9.545* (3.98)	-24.028*** (7.294)
Regime duration (log)	0.065 (0.107)	-0.095 (0.161)
Gini (disposable)	-0.108* (0.047)	-0.102 (0.071)

Note:

Country and year fixed effects included.

All independent variables are one year lagged

Standard errors in parentheses.

* p<0.05, ** p<0.01, *** p<0.001

```

X_mod[, col_idx] <- x_val

sim_out <- sim_function(
  seed      = 666,
  nsim      = 1000,
  coefs     = coefs_vec,
  vcov      = vcov_multinom,
  scenario  = X_mod,           # full observed dataset
  K_minus_1 = K_minus_1
)

ev <- sim_out$ev # [nsim x K]

data.frame(
  x_val    = x_val,
  category = cat_names,
  ev_mean  = colMeans(ev),
  ev_lo    = apply(ev, 2, quantile, 0.025),

```

```

    ev_hi    = apply(ev, 2, quantile, 0.975)
  )
})

results_df <- do.call(rbind, ova_results)

```

```

#results_df

```

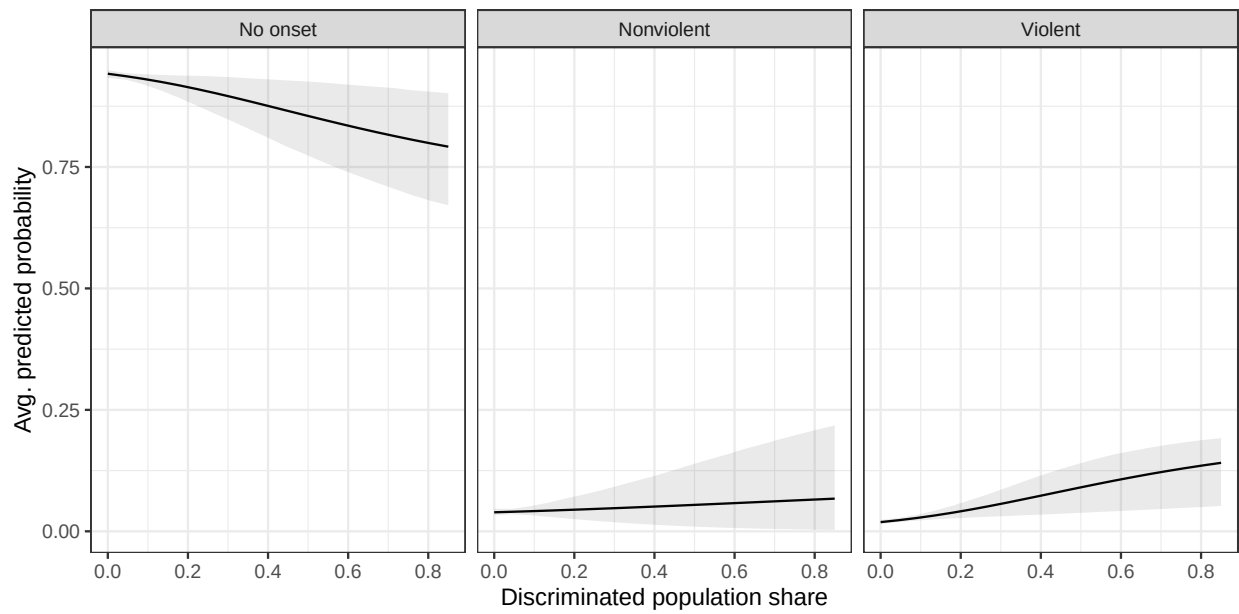


Figure 1: Expected probabilities of event type by the values of discriminated population share

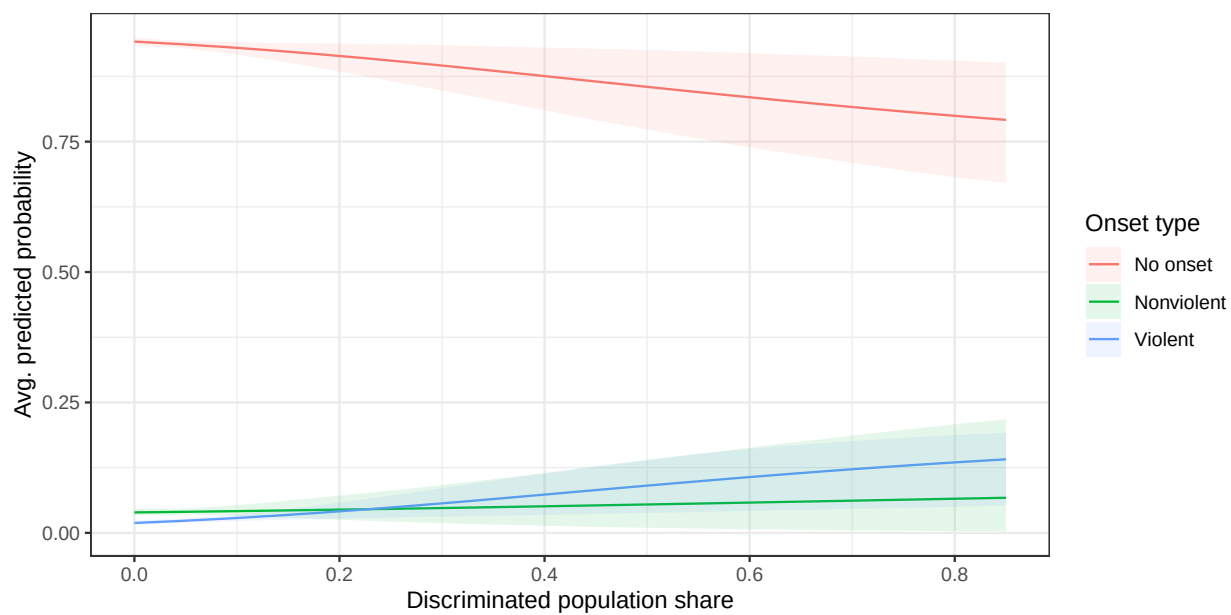


Figure 2: Expected probabilities of event type by the values of discriminated population share - one plot